Motif search and multiple sequence alignment

Task 1. Multiple alignment

Question 1

The best candidate region for the regulatory sequence, in my opinion, is highlighted with blue. The start codons of genes are highlighted with black ATG.

```
VAS14_10979    CAAATATAAACTGTCTTATTTGCGTAAAACTTACGTCATTGAAGTGTAACTCTTGAAGACTAA-ATTTG
VSAL_II0810    ----ACAAAACTGCAATCTTTCTTATATAAAATTGTCATTGTATTGAAATAATCAAATTCTATTATTTG
VSAK1_17757    --TCATTAAACTGCAATCAACCTGCTACAAAACTGTCATCCCAATGAAATATACCAGATTTAA-ATTTG
VCA0602        TGTCATCAGATGGTCACGCATCAGCGATAAAACTGTCACTGCATTGCAATAAAGCCTTGTTAA-TTTTG
VV21669        --TCATCTATTTGTCATCTTCTGGCTATAAATTTGTCATTGGAACGTAACACTCTAAAACTAA-ATTTG
VS_II0401      TGCAATATAACTGTCATTTAGCTACTATAAATTTGTCACTCAACTGCAATATTCTCAGTTTAA-ATTTG
VPA0355        GCAGTTTGACTTGTCATATTTCTGTTACTTAAGCCTCTT-----ATCTTGTTGAGAAAA-TATCTATTG

cons                       *             *         **                       **    ***


VAS14_10979    GTATATACCAATTAGTATG-----AGTTATCACCATGAGTACATATCTCCAAATTGACAACCTTTGTAA
VSAL_II0810    GAGTATACCAAATGGAGAGTTTGAAATGATGGAAAATAAAGCGTACTTAACAATCAATAACGTAATGAA
VSAK1_17757    GTATATACCATTTGGAGAGTAAACGATGTCAGTATCCAAACCTTATCTAGAAATCAACAATGTGGTGAA
VCA0602        GACTATACCAAATTAAAGGTGACTTTTATGACCATGCATCATCCTAATTTAC----------------A
VV21669        GTATATACCATTTGGAGTGTGTGTTATGTCAACTAACCAACCTTACCTAAATATTGAAAATGTTGTGAA
VS_II0401      GTATATACCATGTAGAGAGTGCTGTAGTATGAGCAACCAAACTTATTTGAATATTGAAAACGTTGTAAA
VPA0355        GACTAGGTCAGAATGAAAACAACAGGTGCGGGC--------------CAGTTAGGGACGCAGTTAGGAA

cons           *  **   **                                                           *


VAS14_10979    GCATTTTGGTCAATTCACTGCCTTAAAGAATATCTCCCTCGCTATCAATAAGGGGGAATTTATTTG
VSAL_II0810    AAAATTCGGCCAATTTACGGCCTTAGAACAGATTTCTTTATCTATTGAAAAAGGAGA---------
VSAK1_17757    GCAATTTGGTCAGTTCACGGCATTAAAGCACATTTCCTTGTCCATTAATAAAGGCGA---------
VCA0602        GCAGCACGATCGACCACAAGCGATCAGCAGAAGCTA------------------------------
VV21669        GCAATTCGGTCAGTTCACGGCATTAAAGAATATCTCGCTGGCGATACAAAAAGGCGA---------
VS_II0401      GCAATTTGGCCAATTTACTGCGTTAAAAGACATCTCCCTATCGATAGATAAAGGCGAGTT------
VPA0355        AAATTAAAACCAGCTTGAGACAGCAAATTCACACTGGCGTGGTTTCTCAAGGGCAAAA--------

cons             *       *         *             *
```

Best results aligner: `T-COFFEE, Version_11.00 SCORE=677`

Question 2

I performed MSA alignment using three tools: MUSCLE, ClustalO, Pro-Coffee. Upon results inspection, I found that all those 3 programs mostly agree with each other, but the best looking result with the least amount of gaps was made by Pro-Coffee alignment. MUSCLE was close to Pro-Coffee, and ClustalO gave, in my opinion, the worst result with many gaps. That is, supposedly, because of the method it uses, which tends to propagate gaps once they appear. Pro-Coffee is tuned for short promoter regions so it is good for short conserved regulatory sequences.

The length of the chosen candidate region is 15 nucleotides, although I included a hemi-conserved TA part at the 5’ end, and the sequence of 13 nucleotides without it might also be correct. This sequence appears in all of the upstream regions and doesn’t contain any gaps. It is also located very near the genes start, that is why I have chosen it as a candidate region in question 1.

Question 3

I choose Pro-Coffee aligner with default parameters, because it had the best aligned short conserved region near the beginning of the gene. MUSCLE and ClustalO also found the similar aligned region, but had more gaps. Pro-Coffee also had better alignment for the whole consequent gene region, because it had fewer gaps in it.

MUSCLE region:

AATTTGGsuTATACCA

ClustalO region:

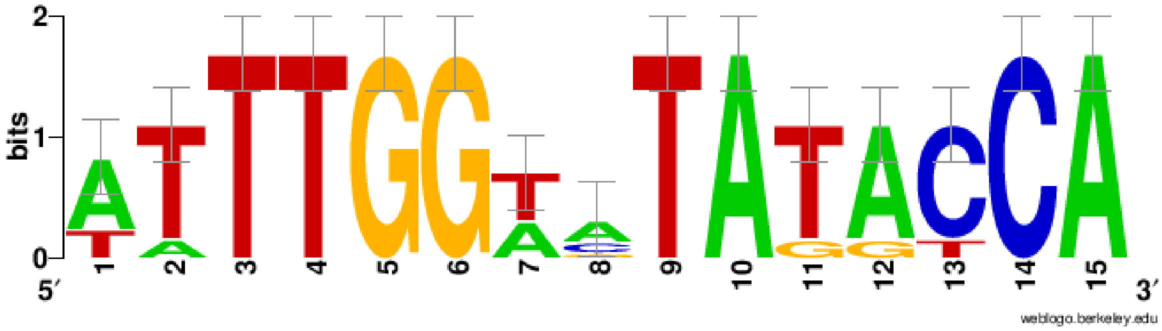
AATTTGGsATATACCA

Pro-Coffee region:

ATTTGGTATATACCA

Task 2. Construction of position weight matrix

Question 4



I used WebLogo tool to obtain the logo.

This logo has a symmetry around 9-10 nucleotides, i.e. 4-9 folds onto 10-15. This further indicates that this is a regulatory region, because transcriptional factors are often dimeric and because of that often bind to palindromic, symmetric DNA regions.

Task 3. Automated search of motifs in the sequences with MEME.

Question 5

Question 6

The logo obtained by MEME is similar to the logo obtained by my manual selection from multiple alignments in most of the nucleotides. The only difference is that I chose a few more nucleotides upstream, while MEME, on the other hand, added a couple nucleotides to the sequence downstream. The logo for the overlapping part of the sequence is completely the same, indicating that MSA was also performed identically.

Question 7

The best match from TomTom:

ID of the motif: MX000004 (Fnr)

Name of transcription factor: Fnr Transcription Factor from Escherichia coli

Database from where the motif is deposited: prodoric_2021.9

E-value of TomTom estimation: 6.72e-01

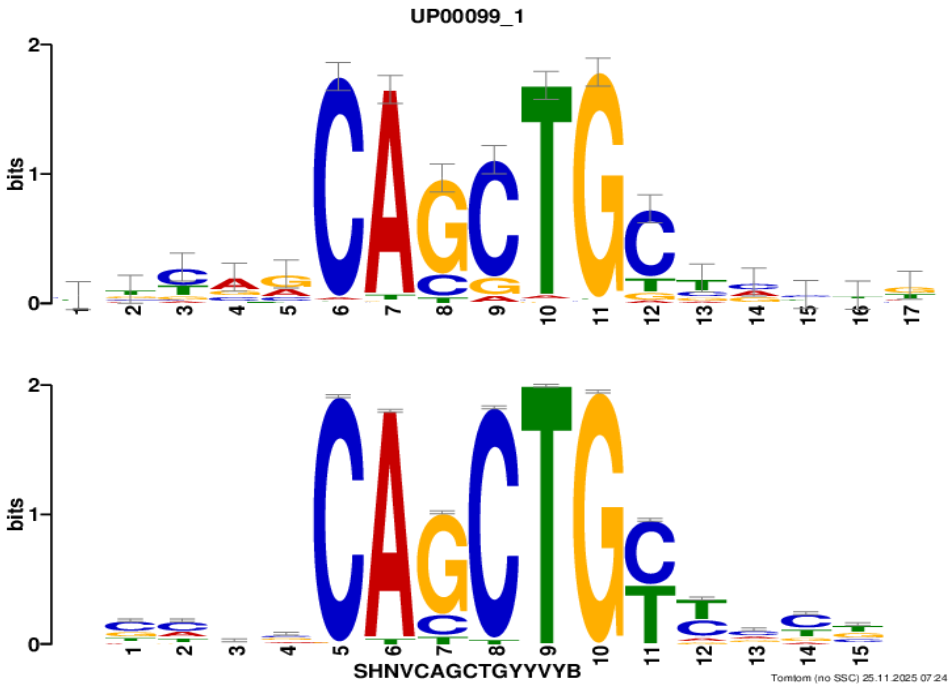
Task 4. Motif search in ChIP-Seq data

Question 8

I submitted my file with peaks to MEME-ChIP service and three hits were found (three motif clusters).

Question 9

Here is the hit with the best E-value of 7.6e-257:



The most similar motif to this hit is UP00099_1 (Ascl2_primary) from uniprobe_mouse database with the E-value equal to 3.01e-04.

According to the CentriMo distribution it is centrally enriched, although with a small dip in the middle, similar to a cowboy hat.

In ChIP-seq for transcriptional factors, antibody pulls down DNA where transcriptional factors bind directly. So, the middle of a peak is a binding site, so motif tends to center.

If it were not central, it could be because of indirect binding, cofactors, noise.

Question 10

JASPAR shows that my ChIP-Seq peaks were obtained for Ascl2 from organism Mus musculus.

UniProbe accession number: UP00099, Uniprot ID: O35885

JASPAR id for a similar hit with a slightly smaller E-value: MA0816.1

There were also some significant hits from Homo sapiens.